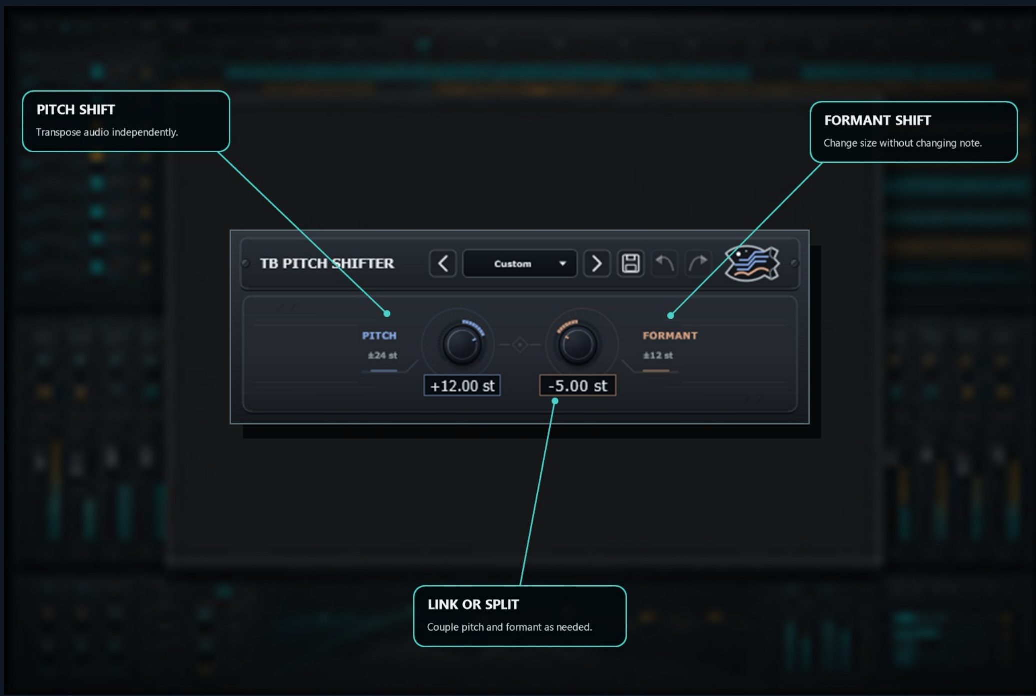
TB Pitch Shifter

DETAILED ENGLISH USER MANUAL

A focused real-time processor with independent Pitch and Formant movement, fixed latency, aligned bypass, and mono/stereo operation.



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Host-visible endpoints documented: 4

1. Product purpose

A focused real-time processor with independent Pitch and Formant movement, fixed latency, aligned bypass, and mono/stereo operation.

Playback-speed changes tie pitch, duration, and perceived source size together. TB Pitch Shifter moves harmonic spacing across four octaves while formants can move independently, allowing octave changes, voice-size transformations, and designed tonal shifts without changing clip duration.

What result it creates

- Pitch shift from -24 to +24 semitones
- Independent formant shift from -12 to +12 semitones
- Automatable mono or stereo transformation
- Aligned bypass and repeatable session recall

Signal flow

Input -> FFT analysis -> harmonic pitch remap -> independent formant-envelope remap -> inverse transform -> latency-aligned output

2. Complete feature guide

Pitch

Moves the musical note and harmonic spacing in semitones. Positive values raise pitch; negative values lower it.

Formant

Moves perceived resonant size independently. Negative values generally sound larger/darker; positive values smaller/brighter.

Independent operation

Set one control to zero to change only the other. Link them manually only when a conventional speed-like size relationship is desired.

Programs

Init, Octave Up, Octave Down, Female to Male, and Male to Female provide direct starting points.

Latency and automation

FFT processing uses fixed reported latency. Automation is smoothed, but very fast large jumps may intentionally reveal spectral transitions.

3. Quick start

1. Start with both controls at zero.

2. Set Pitch for the required musical interval.

3. Adjust Formant to restore natural size or create a deliberate character.

4. Compare with aligned bypass.

5. Automate in musically meaningful transitions and leave headroom for reconstructed peaks.

4. Practical workflows

Natural octave double

1. Set Pitch to +12 or -12.

2. Move Formant slightly in the opposite direction if the source sounds too small or too large.

3. Blend on a parallel track if the dry note must remain.

Expected result: A stable octave layer is created without changing clip duration.

Character transformation

1. Leave Pitch near zero.

2. Move Formant strongly positive or negative.

3. Add downstream EQ only after the character is established.

Expected result: Perceived source size changes while the musical note stays centered.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Dense polyphony and transients may create spectral smearing.
- Extreme shifts can expose phase or formant artifacts.
- Fixed latency should be compensated by the host.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 4 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Bypass VST3 ID 773352680	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Formant VST3 ID 1470039715	-12.00 st to +12.00 st	0.00 st	Automatable	Moves pitch, frequency structure, or perceived resonant size for the named process.
Pitch VST3 ID 106677056	-24.00 st to +24.00 st	0.00 st	Automatable	Moves pitch, frequency structure, or perceived resonant size for the named process.
Program VST3 ID 1886548852	Init, Octave Up, Octave Down, Female to Male, Male to Female	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.